Assimilation Eye — Hardware Guide

How to build the eight-LED ring circuit on a breadboard, for either supported ESP32 board.

- For using and configuring the finished prop, and for troubleshooting, see user-guide.md.
- For building and uploading the firmware, see README.md.

---

1. Supported boards

Both boards carry the same ESP-WROOM-32 module and run the same firmware. They differ only in how many pins they expose and what is printed next to them.

	DOIT ESP32 DEVKIT V1	**ESP32-DevKitC (38-pin)**
Status	**The board in hand**	Also supported
Header pins	30 (15 per side)	38 (19 per side)
Pin labels	`D23`, `D22`, `TX2`, `RX2` …	`IO23`, `IO22`, `IO17`, `IO16` … (some clones print `G23` or `23`)
Controllable onboard LED	Blue, on GPIO 2	None on the genuine board; some clones have one on GPIO 2
5 V pin label	`VIN`	`5V`
BOOT button	Yes	Yes

The board in hand was identified on 2026-10-09: 30 pins, with GPIO 17 and 16 labelled `TX2` and `RX2`. Over USB it reports chip ESP32-D0WD-V3 revision 3.1, 4 MB flash, MAC address `38:18:2b:b3:91:80`.

The wiring in section 7.1 was checked on this board on 2026-10-10 with the bring-up firmware: all eight LEDs light one at a time in the order LED 1 to LED 8, and the blue onboard LED on GPIO 2 blinks under firmware control.

Wire by the printed label, not by counting positions. The positions in this guide are a convenience for finding the label.

---

2. Bill of materials

Qty	**Part**	**Notes**
1	ESP32 board	Either board from section 1
8	Red 5 mm diffused LED	Forward voltage about 1.8–2.2 V, rated 20 mA. Diffused lenses give a wider viewing angle than clear ones
8	220 Ω resistor, ¼ W	Colour bands red, red, brown, (gold). One per LED
1	Breadboard, full size (830 points)	These boards are wide; see section 7
9+	Male-to-male jumper wires	Eight signal wires and one ground wire, plus spares
1	USB cable	USB-C or Micro-USB to match your board. It must be a data cable, not charge-only
1	USB power bank	For the wearable. Mains USB or a PC port is fine on the bench

Optional:

Qty	**Part**	**Notes**
8	100 Ω resistor, ¼ W	Brown, black, brown. The brighter option in section 3
1	Inline USB switch	An accessible way to power the prop off on the cap
2–3	Spare LEDs and resistors	LEDs are easy to damage while bending legs

No transistors, driver chips or external power supply are needed. Section 4 explains why.

---

3. Resistor value

Each LED is driven straight from an ESP32 pin through its own resistor. The resistor sets the current.

Assumptions

- ESP32 pin output when on: 3.3 V
- Red LED forward voltage: 2.0 V typical (range 1.8–2.2 V)
- Target current: about 6 mA, which is clearly visible indoors and gentle on the board

Calculation

```
current = (pin voltage − LED forward voltage) / resistance
        = (3.3 V − 2.0 V) / 220 Ω
        ≈ 5.9 mA per LED
```

	220 Ω (standard)	100 Ω (brighter)
Current per LED, typical	5.9 mA	13 mA
Current per LED, across the 1.8–2.2 V range	5.0–6.8 mA	11–15 mA
All eight LEDs fully on	about 47 mA	about 104 mA
Heat in each resistor	about 8 mW	about 17 mW

Both columns are well within what a ¼ W (250 mW) resistor, a single ESP32 pin and the board's 3.3 V regulator can handle. The comet pattern lights only a few LEDs at once, most of them dimmed, so typical draw is far below the "all eight on" figure. Only the flash pattern at full brightness reaches it.

Which to use: start with 220 Ω. If the ring looks weak under stage lighting, swap all eight for 100 Ω. Do not go below 68 Ω.

Brightness is also adjustable from the web interface, but only downward from what the resistor allows.

---

4. Why no transistor drivers

A transistor driver is needed when a pin cannot supply the voltage or the current a load needs. Neither applies here.

- **Voltage:** a red LED needs about 2.0 V. The pin supplies 3.3 V, which leaves 1.3 V for the resistor to work with.
- **Current per pin:** about 6 mA (or 13 mA with 100 Ω), well under what an ESP32 pin can drive.
- **Total current:** 47–104 mA with everything on, a small part of what the board's regulator supplies.
- **One pin per LED:** no pin carries more than a single LED.

When you would need drivers. Blue, white and "true green" LEDs need 3.0–3.4 V, which is too close to 3.3 V: they light dimly and unevenly. The same applies if you want more than about 15 mA per LED, or several LEDs per pin. In those cases each LED gets one NPN transistor and is powered from the 5 V pin:

```
5 V pin (VIN / 5V) ──[ 150 Ω ]──►|── collector
                                 LED      2N3904 (NPN)
ESP32 pin ──[ 1 kΩ ]──────────────── base
GND ──────────────────────────────── emitter
```

- LED resistor: (5 V − 3.1 V − 0.2 V) / 150 Ω ≈ 11 mA
- Base resistor: 1 kΩ
- The firmware is unchanged: pin high still means LED on.

This arrangement is for reference only. This project uses red LEDs with direct drive.

---

5. Pin assignment

The firmware uses the same eight GPIO numbers on both boards. All eight, and a ground pin, are on one edge of the board.

Ring LED	GPIO	DEVKIT V1 label	DEVKIT V1 position	DevKitC 38-pin label	DevKitC position
LED 1	23	D23	1	IO23	2
LED 2	22	D22	2	IO22	3
LED 3	21	D21	5	IO21	6
LED 4	19	D19	6	IO19	8
LED 5	18	D18	7	IO18	9
LED 6	17	TX2	9	IO17	11
LED 7	16	RX2	10	IO16	12
LED 8	4	D4	11	IO4	13
Ground	—	GND	14	GND	1 or 7

Positions are counted from the antenna end (the end opposite the USB connector) along the edge that carries pin 23.

The full edge, for orientation

DEVKIT V1, 15 pins, antenna end first. Pins used by this project are marked.

Pos	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Label	D23	D22	TX0	RX0	D21	D19	D18	D5	TX2	RX2	D4	D2	D15	GND	3V3
Use	LED 1	LED 2	—	—	LED 3	LED 4	LED 5	—	LED 6	LED 7	LED 8	—	—	Ground	—

DevKitC 38-pin, 19 pins, antenna end first.

Pos	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Label	GND	IO23	IO22	TXD0	RXD0	IO21	GND	IO19	IO18	IO5	IO17	IO16	IO4	IO0	IO2	IO15	SD1	SD0	CLK
Use	Ground	LED 1	LED 2	—	—	LED 3	(Ground)	LED 4	LED 5	—	LED 6	LED 7	LED 8	—	—	—	—	—	—

Pins to leave alone

Pins	Reason
TX0 / RX0 (GPIO 1 and 3)	The USB serial link used for uploading and logs
D5, D2, D15, and GPIO 0 and 12	Read by the chip at power-on to decide how to start. A load on them can stop the board booting or accepting uploads
D2 (GPIO 2)	Also the onboard blue LED on the DEVKIT V1
3V3	A power output, right next to GND on the DEVKIT V1. Nothing in this circuit connects to it. Shorting it to GND can damage the board
SD0–SD3, CLK, CMD (38-pin only)	Wired to the module's internal flash memory
D34, D35, VP, VN	Input only

On the DEVKIT V1, TX2 and RX2 are simply GPIO 17 and 16. They are ordinary outputs here; the second serial port is not used.

---

6. LED polarity and ring layout

6.1 Polarity

An LED only lights when connected the right way round.

	Anode (+)	Cathode (−)
Leg length (new LED)	Longer	Shorter
Rim of the lens	Round	Flat spot
Inside the lens	Smaller metal piece	Larger, cup-shaped piece
Connects to	Resistor, then the ESP32 pin	Ground

A reversed LED does not light and, at these voltages, is not damaged. Turn it round.

6.2 The circuit for one LED

```
ESP32 pin ──[ 220 Ω ]──►|── GND
                     anode  cathode
                    (long)  (short)
```

Every LED has its own resistor. The resistor may go on either side of the LED; this guide puts it on the pin side. All eight cathodes join one common ground, which connects to one GND pin on the board.

6.3 Ring numbering

LED 1 is at the top of the lens. Numbering runs **clockwise as the audience sees it**, looking at the wearer's face. "Clockwise" in the web interface means this direction.

```
        Audience view                    Wearer's view (mirrored)

            LED 1                              LED 1
        LED 8     LED 2                    LED 2     LED 8

      LED 7   lens   LED 3                LED 3   lens   LED 7

        LED 6     LED 4                    LED 4     LED 6
            LED 5                              LED 5
```

LED	Clock position, audience view
1	12:00
2	1:30
3	3:00
4	4:30
5	6:00
6	7:30
7	9:00
8	10:30

On the breadboard, lay the LEDs out in a row in order 1 to 8. The order matters; the circle comes later, on the glasses.

---

7. Wiring

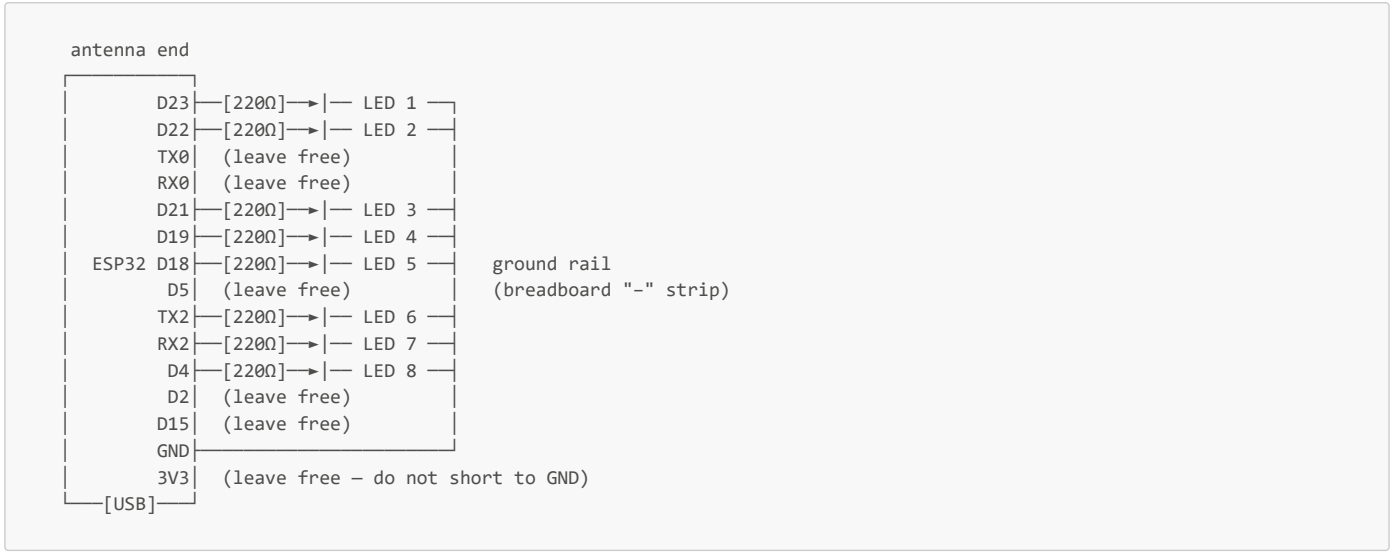
7.1 Connection table — DOIT ESP32 DEVKIT V1

From (board label)	Through	To	Then
D23	220 Ω	LED 1 anode (long leg)	LED 1 cathode → ground rail
D22	220 Ω	LED 2 anode	LED 2 cathode → ground rail
D21	220 Ω	LED 3 anode	LED 3 cathode → ground rail
D19	220 Ω	LED 4 anode	LED 4 cathode → ground rail
D18	220 Ω	LED 5 anode	LED 5 cathode → ground rail
TX2	220 Ω	LED 6 anode	LED 6 cathode → ground rail
RX2	220 Ω	LED 7 anode	LED 7 cathode → ground rail
D4	220 Ω	LED 8 anode	LED 8 cathode → ground rail
GND	wire	ground rail	—

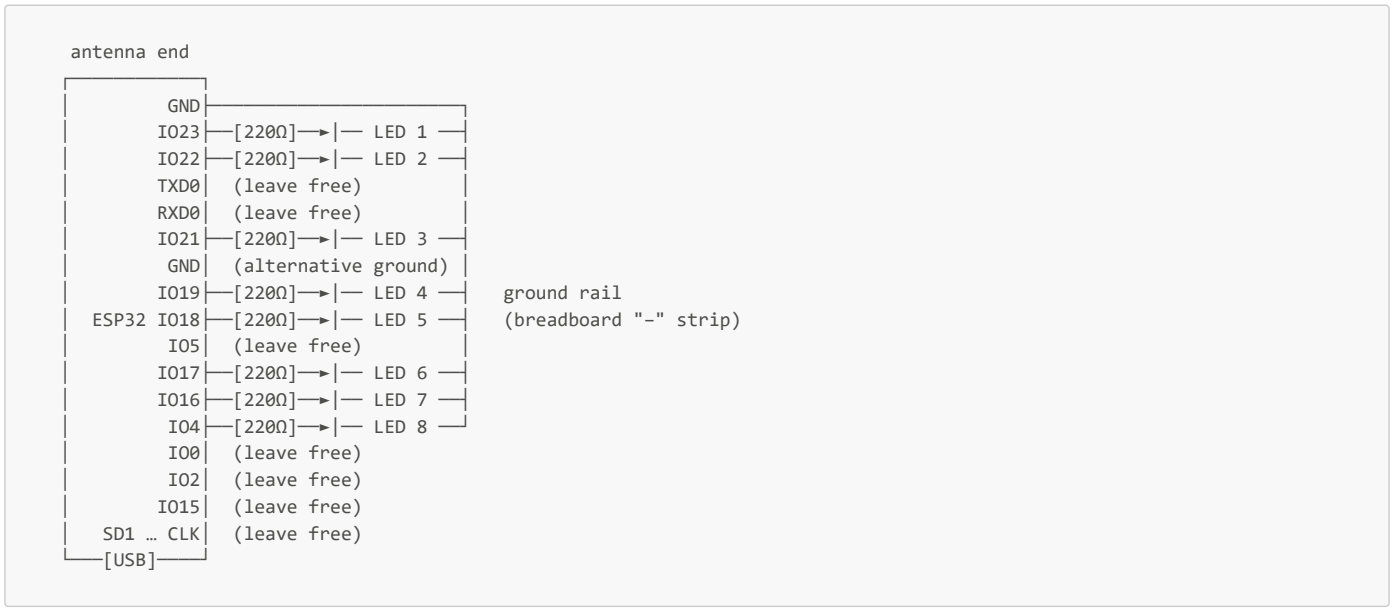
7.2 Connection table — ESP32-DevKitC 38-pin

From (board label)	Through	To	Then
IO23	220 Ω	LED 1 anode (long leg)	LED 1 cathode → ground rail
IO22	220 Ω	LED 2 anode	LED 2 cathode → ground rail
IO21	220 Ω	LED 3 anode	LED 3 cathode → ground rail
IO19	220 Ω	LED 4 anode	LED 4 cathode → ground rail
IO18	220 Ω	LED 5 anode	LED 5 cathode → ground rail
IO17	220 Ω	LED 6 anode	LED 6 cathode → ground rail
IO16	220 Ω	LED 7 anode	LED 7 cathode → ground rail
IO4	220 Ω	LED 8 anode	LED 8 cathode → ground rail
GND (position 1 or 7)	wire	ground rail	—

7.3 Wiring diagram — DEVKIT V1



7.4 Wiring diagram — DevKitC 38-pin



7.5 Power and ground

- **Power:** the board is powered through its USB connector, from a PC, a USB charger or a power bank. Nothing else needs power; the LEDs are fed by the pins.
- **Ground:** one wire from a board `GND` pin to the breadboard's ground rail. All eight LED cathodes go to that rail. This is the common ground.
- **Do not** connect the LEDs to `3V3`, `VIN` or `5V`.
- **Do not** power the board from USB and the `VIN`/`5V` pin at the same time.

---

8. Breadboard assembly

Work with the USB cable **unplugged** until step 9.

How a breadboard is connected. Each numbered row of five holes on one side of the centre channel is joined together. The long strips down the edges, marked `+` and `−`, are joined along their length. On many full-size boards these strips are split in the middle, so keep everything in one half or bridge the gap with a wire.

1. **Seat the board.** Push the ESP32 board into the breadboard across the centre channel, USB connector at one end and overhanging if you like. These boards are wide: you will have only one free hole beside each pin, and on some breadboards only along one edge. Position the board so the free holes are along the **edge that carries pin 23**. Every connection in this circuit is on that edge. Press evenly on both ends; do not rock it.
2. **Make the ground rail.** Run a jumper wire from the board's `GND` pin (on the pin 23 edge) to the `−` strip on that side of the breadboard.
3. **Check the LEDs.** Sort out eight LEDs and note the long leg on each. If legs have been trimmed, use the flat spot on the rim to find the cathode.
4. **Place the LEDs.** In a clear area of the breadboard, away from the ESP32 board, insert the eight LEDs in a line. Give each LED two rows of its own: long leg in one row, short leg in the next. Leave a row or two between LEDs. Keep them in order, LED 1 nearest the antenna end.
5. **Ground the cathodes.** For each LED, run a short jumper from the row holding its **short** leg to the `−` strip.
6. **Add the resistors.** For each LED, put one leg of a 220 Ω resistor in the row holding the LED's **long** leg, and the other leg in an empty row. Resistors have no polarity. Keep bare resistor legs from touching each other.

7. **Connect the signal wires.** For each LED, run a jumper from the free end of its resistor to the board pin in the table in section 7. Do them in order and say the label out loud as you go: `D23`, `D22`, `D21`, `D19`, `D18`, `TX2`, `RX2`, `D4`. Note the three gaps: skip `TX0` and `RX0` after `D22`, and skip `D5` after `D18`.
8. **Inspect before powering.**
   - Eight signal wires, eight resistors, eight LEDs, eight cathode wires, one ground wire.
   - Nothing connected to `3V3`, `VIN`/`5V`, `TX0`, `RX0`, `D5`, `D2` or `D15`.
   - No bare legs touching.
   - Every long leg on the resistor side, every short leg on the ground side.
9. **Power check, no firmware needed.** Plug in the USB cable. The board's red power LED should light. The ring LEDs will stay dark, or do whatever the firmware already on the board tells those pins to do.
10. **Polarity check, no firmware needed.** Unplug one signal wire from its board pin and touch that end to the `3V3` pin. That LED should light at normal brightness. Put the wire back on its pin and repeat for each LED. An LED that stays dark is reversed, or its cathode wire is missing. This is the one time anything touches `3V3`; take care not to brush the neighbouring `GND` pin.

Where the two boards differ

Step	DEVKIT V1 (30-pin)	DevKitC 38-pin
1	15 pins per side	19 pins per side; needs more breadboard length
2	`GND` is second from the USB end, next to `3V3`	`GND` is the first pin at the antenna end; a second `GND` sits between `IO21` and `IO19`
7	Labels `D23` … `D4`, with `TX2` and `RX2` for LEDs 6 and 7	Labels `IO23` … `IO4`, with `IO17` and `IO16` for LEDs 6 and 7
7	Gaps: `TX0`, `RX0`, `D5`	Gaps: `TXD0`, `RXD0`, the middle `GND`, `IO5`
9	Blue onboard LED (GPIO 2) lights when the firmware drives it; confirmed on the board in hand	No controllable onboard LED on the genuine board

9. Initial testing with the firmware

The firmware must be on the board first. README.md covers building and uploading it.

9.1 The power-on self-test

Every time the board is powered or restarted, before anything else:

1. LED 1 lights for a fifth of a second, then LED 2, and so on to LED 8.
2. The whole ring fades up and back down once.
3. The ring starts its pattern.

Watch it two or three times; unplug and replug to repeat it.

What you see	Meaning
Eight LEDs, one at a time, in order 1 to 8	The wiring is correct
A gap where an LED should light	That LED is reversed, or a wire or resistor on it is loose
The right LEDs in the wrong order	Two signal wires are swapped. Compare with section 7
Two LEDs light at the same step	Their wires or legs are touching
All correct, but one is dimmer in the fade	Wrong resistor value on that LED
Nothing lights, red power light on	The ground wire is missing, or BOOT was held while powering on

9.2 What the serial console shows

Optional, with the board on a PC. Open the console as described in the README. A healthy start looks like this, with your own version, addresses and times:

```
=== Assimilation Eye ===
Firmware 0.8.0, built 2026-10-10 17:44
Chip ESP32-D0WD-V3 rev 3.1, 2 core(s), flash 4 MB. Last start: power on.
Firmware 1192272 bytes of a 1966080-byte slot
LED ring: 8 outputs at 5000 Hz, 12-bit — ready
Onboard LED: ready
Self-test: LED 1 -> LED 8, then fade.
Self-test done.
Settings: restored from flash
LED defaults: from this build's led-defaults.json
Animation: 3 patterns, 5 ms tick, pattern "blend" — running
[auth] Sign-in user "admin", DEFAULT PASSWORD still in use. A password check takes 227 ms.
[wifi] Access point "AssimilationEye" is up. Interface address: http://192.168.4.1
```

```
[wifi] No network saved. Use the Wi-Fi screen to join one.
[api] Listening on port 80.
```

A line containing `FAILED` names the part that did not start.

9.3 Checking each LED from the browser

1. Connect and sign in (user guide, section 3).
2. On the Eye screen choose **Synchronized flash**, set **Minimum brightness** and **Maximum brightness** to 100% and **Overall brightness** to 100%. All eight LEDs are now fully and steadily on. Compare their brightness.
3. Choose **Rotating comet**, set **Tail length** to 0 and **Speed** to its slowest. A single LED travels round. Confirm it visits every position in order.
4. Tap **Reset LED settings** to put everything back.

9.4 Before moving on

- ☐ The self-test is correct three times in a row
- ☐ All eight LEDs are equally bright in step 2 above
- ☐ The board is not warm to the touch after five minutes at full brightness
- ☐ The prop runs for 15 minutes from the power bank you will use, without cutting out

---

10. Troubleshooting

Hardware and software troubleshooting are together in the user guide, laid out as symptom, likely cause and fix:

- User guide, section 10.2 — Hardware: dark, dim, flickering or out-of-order LEDs, a board that will not power or is not detected, a power bank that cuts out.
- User guide, section 10.1 — Connecting, signing in and updating.

Checks that need no firmware are steps 9 and 10 of section 8.

---

11. Transferring to the wearable

Do this only after section 9.4 is complete on the breadboard. A fault is far easier to find there than on a cap.

11.1 Extra parts

Part	Notes
Thin stranded wire, about 26–30 AWG, ideally in several colours	Solid breadboard wire breaks when flexed. Nine conductors are needed; ribbon cable works well
Heat-shrink tubing, 1.5–2 mm	To insulate every joint. Electrical tape comes loose with warmth
Soldering iron and solder	
A ring to hold the LEDs	Stiff card, craft foam or thin plastic, cut to fit around the lens, with eight holes
Hot glue or epoxy	To fix LEDs and relieve strain on wires
Small cable ties, Velcro or fabric tape	To route wires and mount the board and power bank
Short USB cable, and an inline USB switch	The cable from the power bank to the board, with the off switch in it

11.2 Plan the layout

- **LED order.** Mark eight evenly spaced positions around the lens. LED 1 is at the top; numbering runs clockwise **as the audience sees it**, as in section 6.3. Working from behind the glasses, that is counter-clockwise. Write the numbers on the back of the ring before fitting anything.
- **Which lens.** Either. The firmware does not care.
- **Board.** On the side or back of the cap, where the USB socket is reachable and the **BOOT button is not pressed by anything**. A pressed BOOT button at power-on stops the board starting, and held for ten seconds it erases the settings.
- **Power bank.** In a pocket, or at the back of the cap as a counterweight.
- **Wires.** Along the arm of the glasses and up to the cap. Leave slack where the glasses meet the cap so they can be put on and taken off.
- **Vision.** Nothing in front of the lens. Keep the ring on the frame, and point the LEDs outward, away from the eye.

11.3 Build the ring

1. Push the eight LEDs through the holes from the front, all turned the same way, so that every **short leg (cathode)** points to the same side, for example toward the centre.
2. Fix each LED with a small dab of glue at the back.
3. **Common ground.** Bend each short leg over to reach its neighbour and solder them into a ring, or run one bare wire around and solder each short leg to it. Solder one wire to this ring: this is the ground wire.

4. **Resistors.** Trim each long leg to about 5 mm and solder a 220 Ω resistor to it. Slide heat-shrink over the joint and the resistor body.
5. **Signal wires.** Solder one wire to the free end of each resistor. Use a different colour for each, or label both ends 1 to 8 with tape.
6. Insulate everything. No bare metal on one LED's signal side may be able to touch the ground ring or another LED's signal.

The resistors can instead go at the board end of the wires if there is no room at the ring. The circuit is the same.

Solder quickly: more than about three seconds of heat on a leg can damage an LED. Hold the leg with pliers between the joint and the LED to draw heat away.

11.4 Connect to the board

Solder the wires to the board's pins, or use female jumper leads pushed onto the pins and secured with a spot of glue.

Wire	Board label, DEVKIT V1	Board label, DevKitC 38-pin
LED 1	`D23`	`IO23`
LED 2	`D22`	`IO22`
LED 3	`D21`	`IO21`
LED 4	`D19`	`IO19`
LED 5	`D18`	`IO18`
LED 6	`TX2`	`IO17`
LED 7	`RX2`	`IO16`
LED 8	`D4`	`IO4`
Ground	`GND`	`GND`

If two wires end up swapped, they can be corrected without soldering: reorder the pin list `RING_LED_PINS` in `include/HardwareConfig.h` and rebuild the firmware. If the whole ring runs backwards, simply choose the other direction in the browser.

11.5 Test before mounting

Power the board and watch the self-test with the ring lying on the table. Check the order against the numbers written on the back, **looking from the front**. Fix any fault now.

Then flex every wire gently near its joints while the ring is fully on (section 9.3, step 2). An LED that flickers has a weak joint.

11.6 Mount

1. Fix the ring to the frame around the lens.
2. Run the wires along the arm. Tie them down every few centimetres.
3. **Strain relief.** Fix each wire bundle to something solid within a centimetre or two of its joints, at the ring and at the board, so that movement bends the wire and not the solder.
4. Mount the board on a non-conductive surface. Cover its underside with tape or put it in a small case: the pins are bare metal, and sweat, foil or a metal cap fitting can short them.
5. Mount the power bank and connect it through the inline switch.

11.7 Switching the prop off

The firmware has no off button, by design: the prop is switched off by removing power.

- **Inline USB switch** between the power bank and the board, placed where your hand finds it without looking. This is the recommended way.
- Or unplug the USB cable.
- A power bank's own button usually cannot switch the output off.

After changing a setting, wait two seconds before switching off so the change is stored. Switching off at any other time is harmless.

To darken the ring while keeping the prop powered and reachable, use **Stop the ring** in the browser.

11.8 Final check on the wearable

- [ ] Self-test correct with the prop being worn
- [ ] No flicker when the head turns, nods, or the glasses are taken off and put on
- [ ] Nothing presses the BOOT button
- [ ] Nothing obstructs vision; no LED shines into the eye
- [ ] The off switch can be reached while wearing it
- [ ] Runs 15 minutes on the power bank, worn, without cutting out and without anything getting warm
- [ ] The user guide's pre-demo checklist is done